# RITIK GUPTA

iOS Developer | Swift, SwiftUI | Application Architecture & Secure Systems | 4+ Years

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## PROFESSIONAL SUMMARY

iOS Developer with 4+ years of experience designing, building, and shipping secure, market-ready applications using Swift and SwiftUI. Focused on application-to-server communication design — data flow, state management, and secure API integration — combined with MVVM/Coordinator architecture and hardware-backed security (Secure Element, Keychain). Strong track record of taking a design from architecture through production-ready implementation. Additional experience extending products to Android and working with blockchain-based systems.

## TECHNICAL SKILLS

Languages: Swift, Python, Kotlin, Java, JavaScript

iOS & Frameworks: SwiftUI, UIKit, Combine, SwiftData, CoreData, CoreBluetooth, XCTest, Swift Package Manager

Architecture: Application-Server Communication Design, Low-Level Design (LLD), MVVM, MVC, Coordinator Pattern, Protocol-Oriented Programming, Dependency Injection

Security: Secure Key Management, Encryption/Decryption, Secure Element, Keychain Services, JavaCard, GlobalPlatform SCP, OAuth 2.0, SSL/TLS

Networking: URLSession, REST APIs, JSON-RPC, WebSocket, Web3

Standards: ETSI, GSMA, GlobalPlatform, CCC Digital Key

Also: Android (Kotlin, Java, Room), Solidity/Blockchain

Tools & Platforms: Xcode, Git, CI/CD, Instruments, JIRA, Figma, TestFlight, App Store Connect, Claude AI, ChatGPT, GitHub Copilot

## PROFESSIONAL EXPERIENCE

**IDEMIA — Software Engineer** Aug 2022 – Present

*Noida, India*

- Designed application-to-server communication architecture (LLD) for a production iOS crypto wallet application, defining data flow, state management, and secure integration across multiple backend microservices.
- Led development of secure, market-ready iOS applications using Swift, SwiftUI, and the MVVM/Coordinator pattern, taking features from design through to production release for enterprise clients.
- Architected cryptographic solutions including secure key management, encryption/decryption, and MAC generation, leveraging Secure Element and iOS Keychain for enterprise-grade security.
- Built automated testing frameworks using Python and XCTest, achieving 90% code coverage across security-critical modules and UI components.
- Mentored three junior developers on iOS architecture patterns, secure coding practices, and GlobalPlatform/ETSI/GSMA standards compliance.
- Collaborated with Qualcomm on processor-integrated SIM technology, contributing to secure embedded-to-mobile system design.

**IDEMIA — Associate Software Engineer** Feb 2022 – Jul 2022

*Noida, India*

- Mastered core SIM card OS components including Framework, Toolkit, and GlobalPlatform.
- Implemented cryptographic features including secure key management and encryption protocols.
- Developed features compliant with ETSI/3GPP standards: ETSI 102 221 / 3GPP TS 31.101 (UICC Commands), ETSI 102 223 (UICC Application Toolkit), ETSI 102 225/226 (Secured Packet / OTA).
- Contributed to the codebase following secure coding practices and industry standards.

## KEY PROJECTS

**CryptoWallet iOS App & SDK — IDEMIA** Jun 2025 – Present

- Architected and developed a comprehensive iOS SDK to interface with hardware wallets, abstracting complex blockchain interactions for modular, plug-and-play use across multiple applications.
- Designed the application-to-server communication architecture (LLD), reducing network calls by 40% through request batching and unified data flow design.
- Designed and implemented a responsive iOS UI using SwiftUI, streamlining the user experience for asset management and transaction flows.
- Engineered multi-chain support (Ethereum, Polygon, BSC) with a unified transaction interface and real-time validation.
- Integrated JSON-RPC and Web3 protocols to facilitate secure communication between the mobile application and blockchain nodes.
- Extended the product to Android using Kotlin and Java, implementing local persistence with Room and native XML-based UI, achieving full cross-platform parity.

**Digital Car Key — Secure Element Applet | Client: Infineon Technologies** Aug 2024 – Apr 2025

- Developed the JavaCard Applet for the Digital Car Key secure element, implementing secure key storage and cryptographic operations in compliance with the CCC (Car Connectivity Consortium) Digital Key specification.
- Implemented GlobalPlatform SCP (Secure Channel Protocol) for authenticated, encrypted communication between the secure element and the host device.
- Applied GlobalPlatform standards for applet lifecycle management, including installation, personalization, and secure channel establishment.
- Handled key generation, mutual authentication, and key lifecycle management at the applet level to support secure vehicle access.

**Blockchain-Based Identity Infrastructure for AI Agents — Independent Research** Mar 2026 – Present

- Researching a deterministic, self-custodial identity standard for AI agents built on established HD wallet key-derivation principles, extending existing agent-identity frameworks in the Web3 ecosystem.
- Designing a companion payment credentialing layer to bridge verified on-chain agent identity with traditional (Web2) payment infrastructure.
- Patent application in preparation; technical details withheld pending filing.

## EDUCATION

Bachelor of Technology, Computer Science — Lovely Professional University, Phagwara (2018 – 2022)